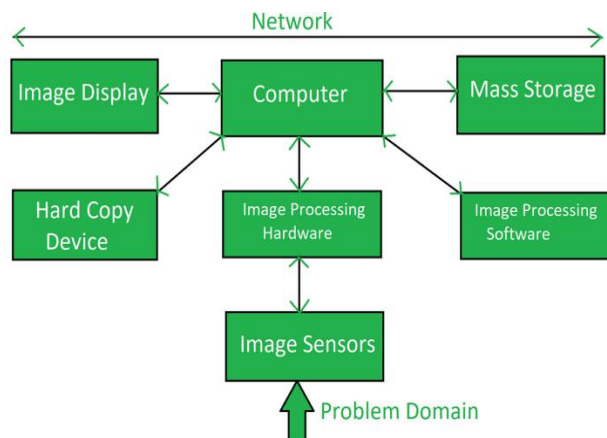


Digital Image Processing



Dr. Md. Aynal Haque

Professor, EEE, BUET

Email: aynal@eee.buet.ac.bd

Cell: 018 1763 8600



Topic 1

Introduction to Image Processing



Topic Outline

- Introduction to Image Processing
- Historical Perspective
- Principal Areas of Application
- Approaches or Steps of Image Processing
- Components of Image Processing Unit



Image

An image is a visual representation of something. Images can take the form of photographs, graphics and individual video frames. For this purpose, an image is a picture that was created or copied and stored in electronic form.

An image is defined as a two-dimensional function, $F(x,y)$, where x and y are spatial coordinates, and the amplitude of F at any pair of coordinates (x,y) is called the **intensity** of that image at that point. When x,y , and amplitude values of F are finite, we call it a **digital image**.

In other words, an image can be defined by a two-dimensional array specifically arranged in rows and columns.

Digital image is a binary representation of visual data. It is composed of a finite number of elements, each of which have a particular value at a particular location. These elements are referred to as *picture elements*, *image elements*, or *pixels*. A *Pixel* is most widely used to denote the elements of a Digital Image.



Types of Image

Binary Image: The binary image as its name suggests, contain only two pixel elements, 0 & 1, where 0 refers to black and 1 refers to white. This image is also known as Monochrome.

Black and White Image: The image which consist of only black and white color.

8-bit Color Format: It is the most famous image format. It has 256 different shades of colors in it and commonly known as Grayscale Image. In this format, 0 stands for Black, and 255 stands for white, and 127 stands for gray.

16-bit Color Format: It is a color image format. It has 65,536 different colors in it. It is also known as High Color Format. In this format the distribution of color is not as same as Grayscale image.

A 16 bit format is actually divided into three further formats which are Red, Green and Blue. That famous RGB format.



Image Processing

Image processing describes the process of transforming an image and executing specific operations to obtain useful information from it. Image processing systems often treat images as 2D signals when applying some predetermined **signal processing** approaches.

Types of image processing include the following:

- **Pattern recognition** to measure various patterns around objects in an image;
- **Image Recognition** to detect or differentiate objects within an image;
- **Image Retrieval** to browse or search an extensive image database for an image like the original image;
- **Image Sharpening and Restoration** to create an enhanced image from the original image; and
- **Image Visualization** to identify objects not visible in an image.



Digital Image Processing

Digital Image Processing means processing digital image by means of a digital computer. We can also say that it is a use of computer algorithms, in order to get enhanced image either to extract some useful information. So, it is the use of algorithms and mathematical models to process and analyze digital images. The goal of digital image processing is to enhance the quality of images, extract meaningful information from images, and automate image-based tasks.

Steps of Image Processing:

- Importing the image via image acquisition tools
- Analyzing and manipulating the image
- Output in which result can be altered image or a report which is based on analyzing that image



DIP: Basic Steps

Image Acquisition: This involves capturing the image using a sensor, such as a camera, scanner, or other device.

Image Enhancement: This step improves the visual quality of the image by adjusting brightness, contrast, and other properties.

Image Restoration: This stage aims to repair or recover degraded images, such as those affected by noise or blur.

Color Image Processing: This deals with manipulating color images, including color space conversions and color corrections.

Wavelets and Multiresolution Processing: This technique allows images to be represented at different resolution levels, enabling efficient analysis and processing.

Image Compression: This step reduces the storage space and transmission bandwidth of images while maintaining acceptable quality.

Morphological Processing: This involves manipulating image shapes and structures using set operations like erosion, dilation, opening, and closing.

Image Segmentation: This stage separates an image into different regions based on intensity, color, or other characteristics.

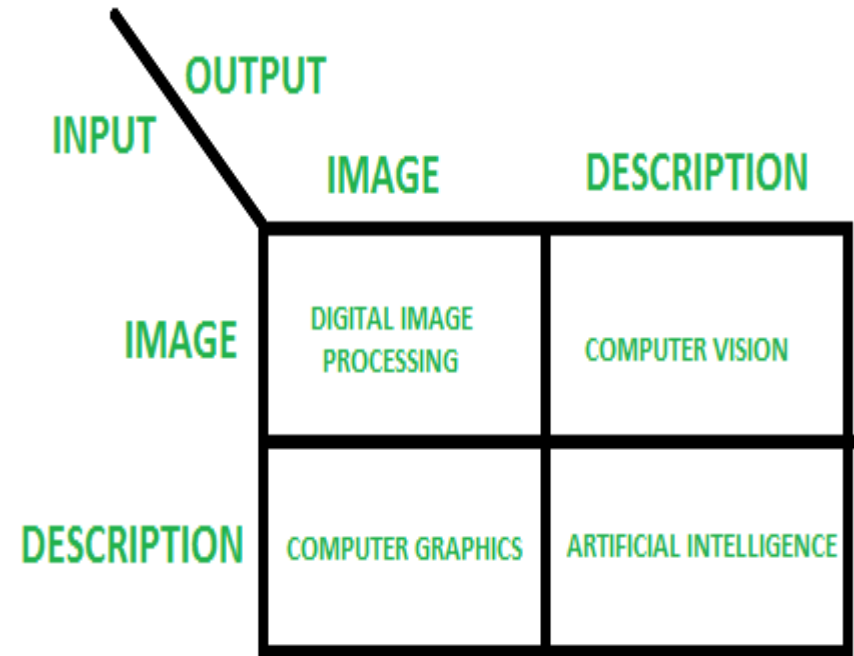
Object Recognition: This involves identifying and classifying objects within an image based on their shape, texture, and other features.

Image Representation and Description: This stage focuses on transforming the image into a form suitable for computer processing, often involving boundary or region representation.



DIP Related Fields

- ❖ If input is an image and we get out image as a output, then it is termed as **Digital Image Processing**
- ❖ If input is an image and we get some kind of information or description as a output, then it is termed as **Computer Vision**
- ❖ If input is some description or code and we get image as an output, then it is termed as **Computer Graphics**
- ❖ If input is description or some keywords or some code and we get description or some keywords as a output, then it is termed as **Artificial Intelligence**





DIP: Advantages

- ✓ **Improved image quality:** Digital image processing algorithms can improve the visual quality of images, making them clearer, sharper, and more informative.
- ✓ **Automated image-based tasks:** Digital image processing can automate many image-based tasks, such as object recognition, pattern detection, and measurement.
- ✓ **Increased efficiency:** Digital image processing algorithms can process images much faster than humans, making it possible to analyze large amounts of data in a short amount of time.
- ✓ **Increased accuracy:** Digital image processing algorithms can provide more accurate results than humans, especially for tasks that require precise measurements or quantitative analysis.



DIP: Limitations

- ❖ **High computational cost:** Some digital image processing algorithms are computationally intensive and require significant computational resources.
- ❖ **Limited interpretability:** Some digital image processing algorithms may produce results that are difficult for humans to interpret, especially for complex or sophisticated algorithms.
- ❖ **Dependence on quality of input:** The quality of the output of digital image processing algorithms is highly dependent on the quality of the input images. Poor quality input images can result in poor quality output.
- ❖ **Limitations of algorithms:** Digital image processing algorithms have limitations, such as the difficulty of recognizing objects in cluttered or poorly lit scenes, or the inability to recognize objects with significant deformations or occlusions.
- ❖ **Dependence on good training data:** The performance of many digital image processing algorithms is dependent on the quality of the training data used to develop the algorithms. Poor quality training data can result in poor performance of the algorithm.



Digital Image

Digital Image is defined in terms of pixel intensity $f(x,y)$

$$x \in Z \quad y \in Z \quad \text{where } f \in Z \text{ or } f \in R$$

Any visual signal is treated as Image, e.g., signal generated from visual band, responses from EM wave, ultrasound, electron microscopy, and computer generated

□ Related areas – Image Analysis and Computer Vision



Inference from Images
Recognition of objects



Emulate HVS – A branch of AI
Take action through robotics



Scope

General Idea about Digital Image Processing



- Low-level Image Processing – Denoising, Enhancement, Restoration
- Mid-level Image Processing – Segmentation, Description of Objects, Classification
- High-level Image Processing – Recognition, Cognitive functions, making sense as HVS



History

Image Communication – 1920s
: London to NY by Newspaper



Image Printed by Telegraphic Printer at the
receiver end – 1921

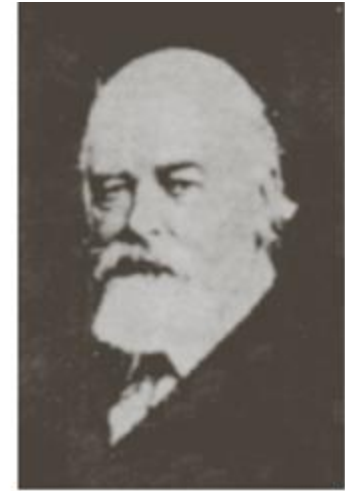


Image Printed by Photographic Tape Printer at
the receiver end – 1922



Improved quality



History



Image Printed by 15-tone Printer at the receiver end – 1929



History

Break through of Image Processing Happens due to

- Computer Technology
 - Memory holds program and data
 - Conditional branching
- Transistor (Bell Laboratory) – 1948
- Programming language (COBOL, FORTRAN) - 1950-60
- IC Technology (Texas Ins.) – 1958
- OS – 1960
- Microprocessor (Intel) – 1970
- PC (IBM) – 1981
- VLSI to ULSI – 1980-2000



History

First Image Processing Task – Restoration of Image Taken from Moon from Ranger 7 on July 1964

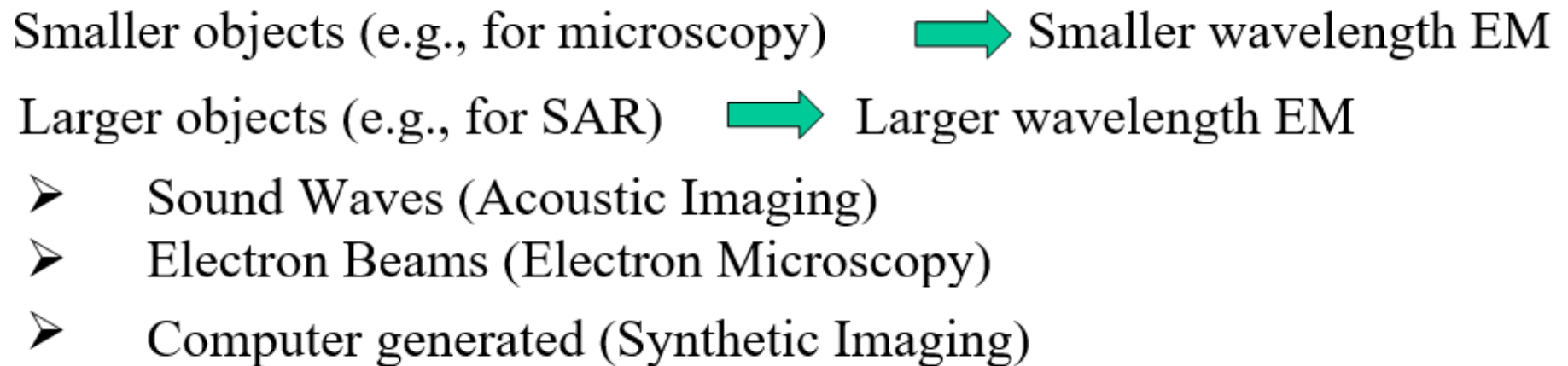




History

Inventions on imaging technology those make revolution for image processing

- ✚ X-ray (1895) –
1901 Noble Prize in Physics
- ✚ CT imaging (1970, 3D rendition of an object using X-ray) –
1979 Noble Prize in Medicine
- ✚ MR imaging (1950, magnetization gradient) –
2003 Noble Prize in Medicine
- ✚ CCD imaging (1969, transfer charge along the surface of a semiconductor from one storage capacitor to the next) –
2009 Noble Prize in Physics





Typical Bands for Remote Sensing

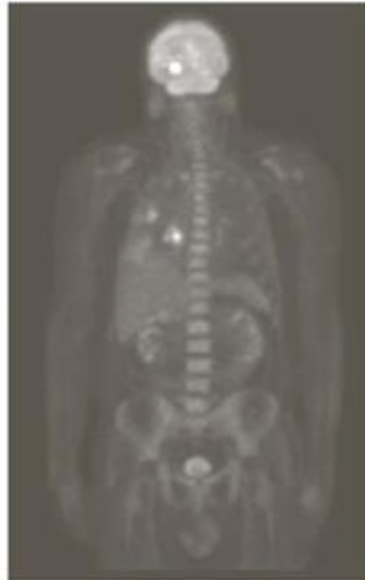
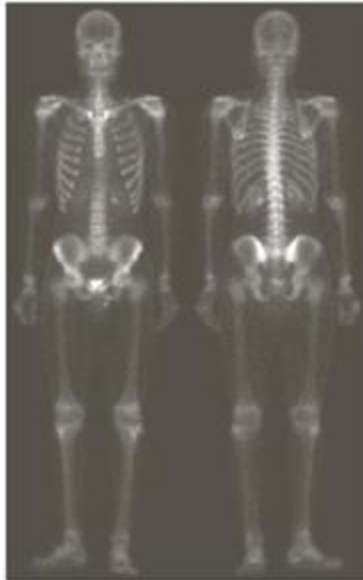
Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.52–0.60	Good for measuring plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.76–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content of soil and vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Middle infrared	2.08–2.35	Mineral mapping



Applications

Gamma-ray imaging

Injects radioactive isotope that emits gamma rays as it decays



A B

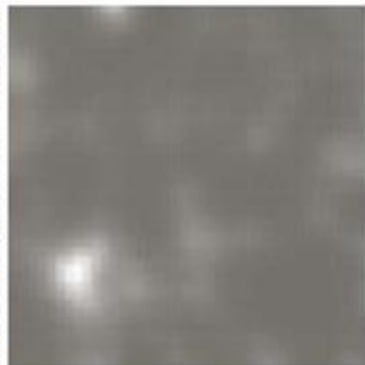
A. Bone Scan

C D

B. PET image

C. Cygnus loop

D. Reactor Valve



PET imaging injects radioactive isotope that emits positron as it decays. Positron meets with electron to give off gamma rays.



Applications

X-ray imaging

At a heated condition cathode released electrons for anode.
Electron hits nucleus to radiate X-rays)



A D

A. Chest X-ray

D. Circuit Board

B E

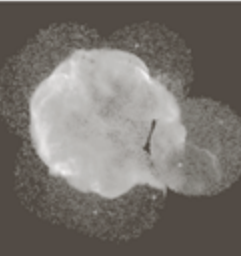
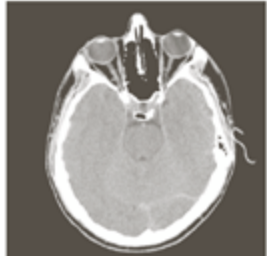
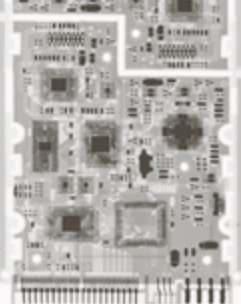
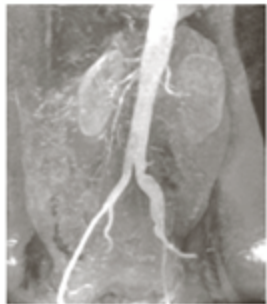
B. Angiogram

E. Circuit Board

C F

C. Head CT

F. Cygnus Loop



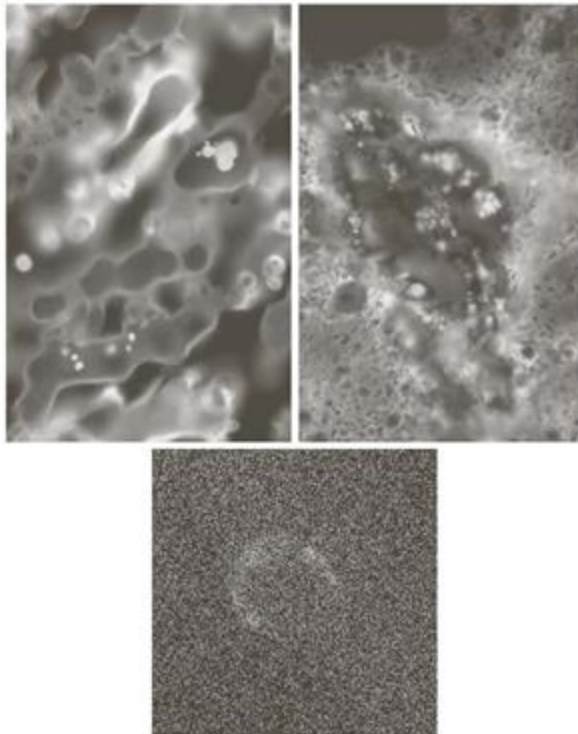
- X-ray contrast medium is used for angiogram
- CT image produce 3D rendition



Applications

Ultraviolet Imaging

Photon of an ultraviolet radiation collides with atom of a fluorescent material, an excited high-energy electron releases visible (red) light.



- A B A. Normal Corn
C B. Smut Corn
 C. Cygnus loop

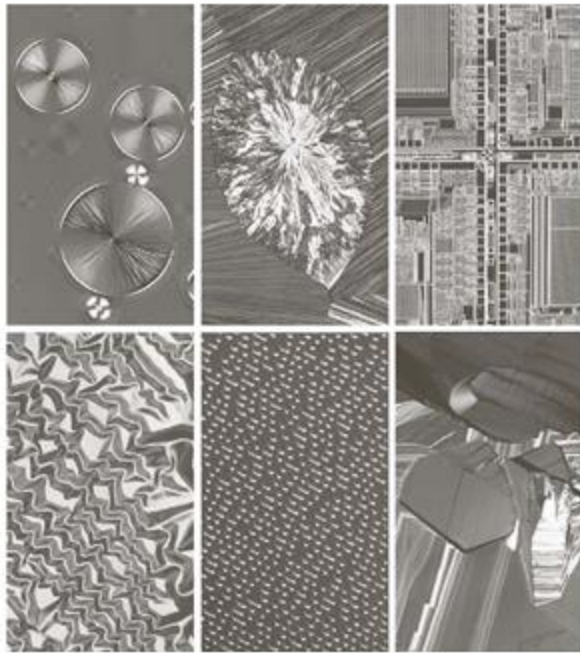
Secondary fluorescence may also be used instead of primary fluorescence



Applications

Visible and Infrared Imaging

Microscopic applications use low wavelength EM wave. CCD sensor for visible range. IR or heat sensor for infrared sensing.



A B C

D E F

A. Taxol B. Cholesterol C. μ -processor

D. Thin Film E. CD F. Semiconductor

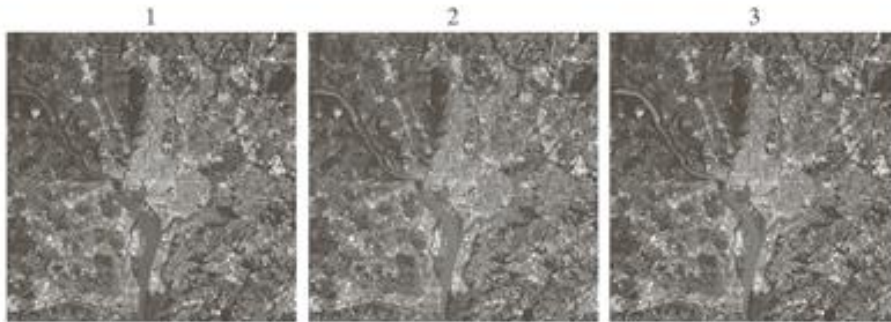
□ Microscopy: zoom ranges 40 $\times$ to 1750 $\times$



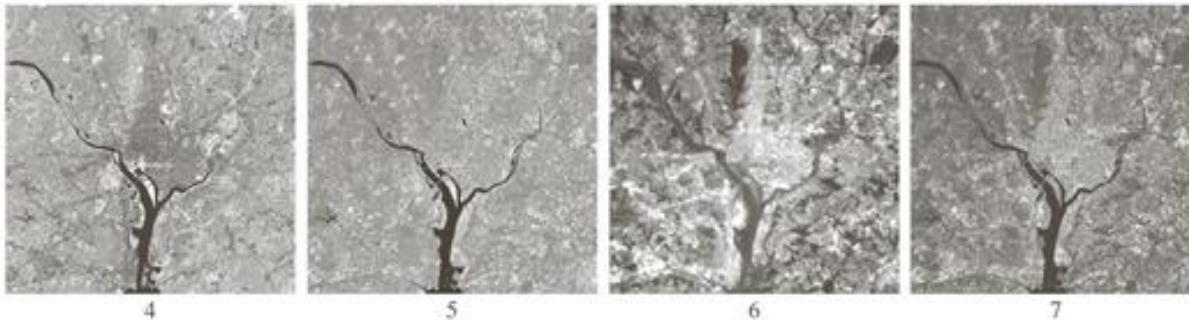
Applications

Visible and Infrared Imaging

Multiple spectrums of EM are used for remote sensing



1-6: Satellite images –
Washington DC, USA



Remote sensing observes – population growth, and shift patterns, floods, deforestation, other environmental changes



Applications

Visible and Infrared Imaging

Multiple spectrums of EM are used for meteorology



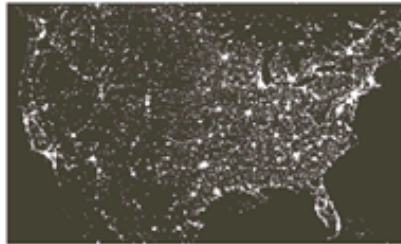
Satellite image -
Hurricane Andrew



Applications

Infrared Imaging

Night time light – determines economic development in terms of energy consumption



Night-time
Satellite image –
North and South
Americas



Applications

Infrared Imaging (Night time light)

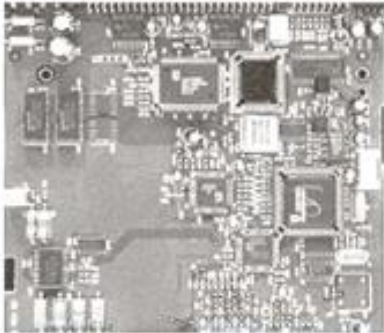


Night-time Satellite
image –
Africa, Europe, Asia,
Australia



Applications

Visible Imaging (Manufacturing Industry)



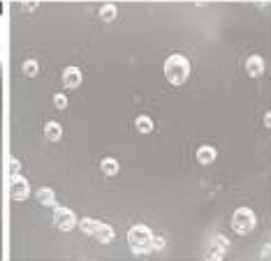
A B
C D
E F

A. Circuit
Board
C. Bottles
E. Cereal

B. Pill
Package

D. Plastic

F. Intraocular
implant





Applications

Visible Imaging (Forensic and Security)



- A B A. Finger Print
C
D B. Paper Currency
C. & D. License Plate Recognition

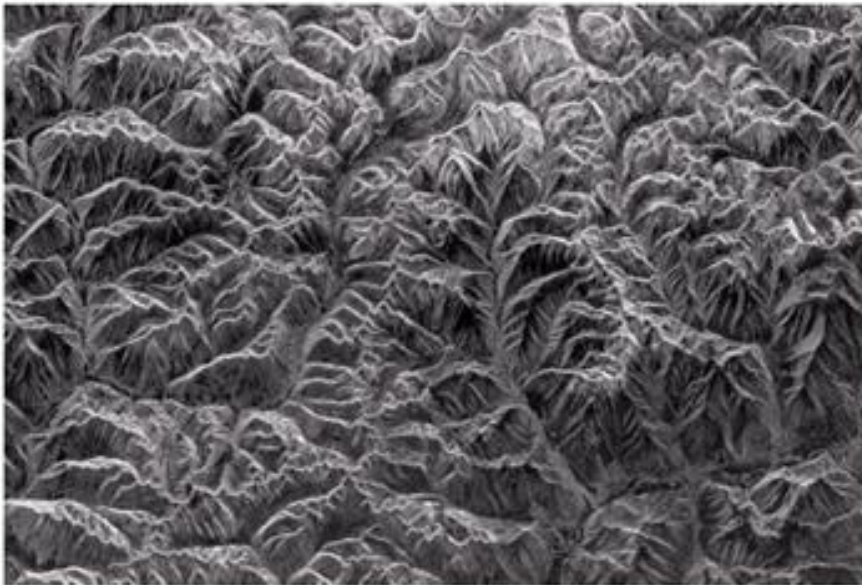




Applications

Microwave Imaging

Used in Radar, independent on weather condition



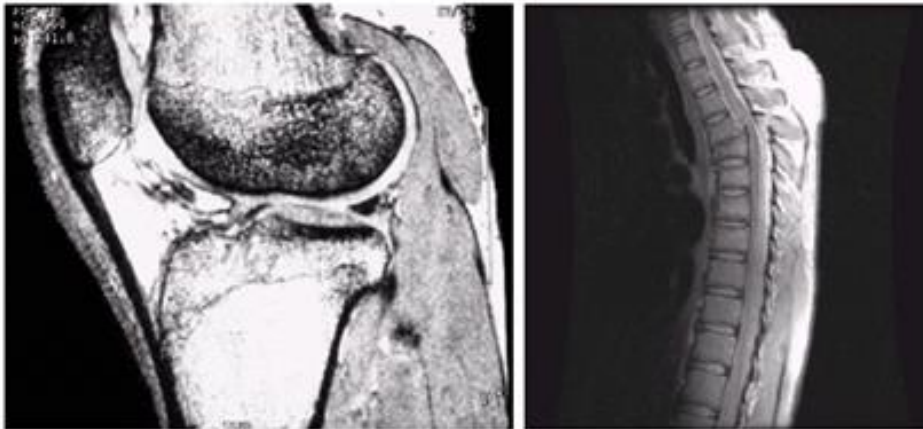
Space-borne radar
image of Tibet



Applications

Radio wave Imaging

Used in MRI – in a strong magnetic field, radio waves in terms of pulses pass through patient. Strength and location of responding pulses are detected



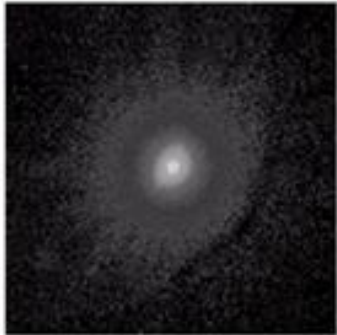
A B A. Knee MRI
 B. Spine MRI



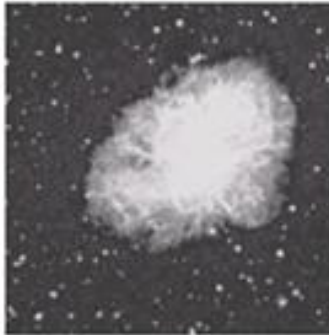
Comparison



Gamma



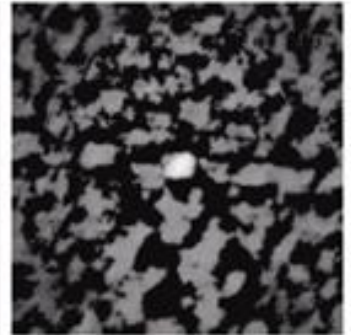
X-ray



Optical



Infrared



Radio

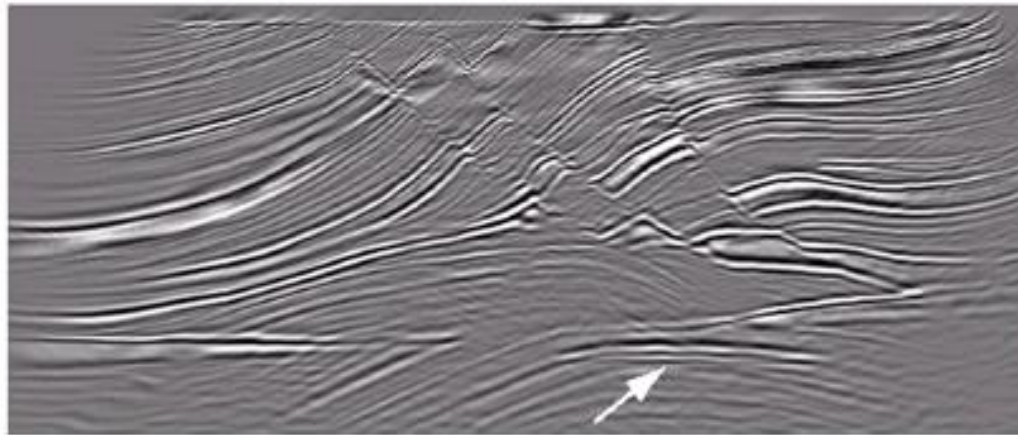
Images of Crab Pulsar



Applications

Low frequency (~ 100 Hz) sound wave Imaging

Used in geological exploration, 3D map of ocean bed, etc.



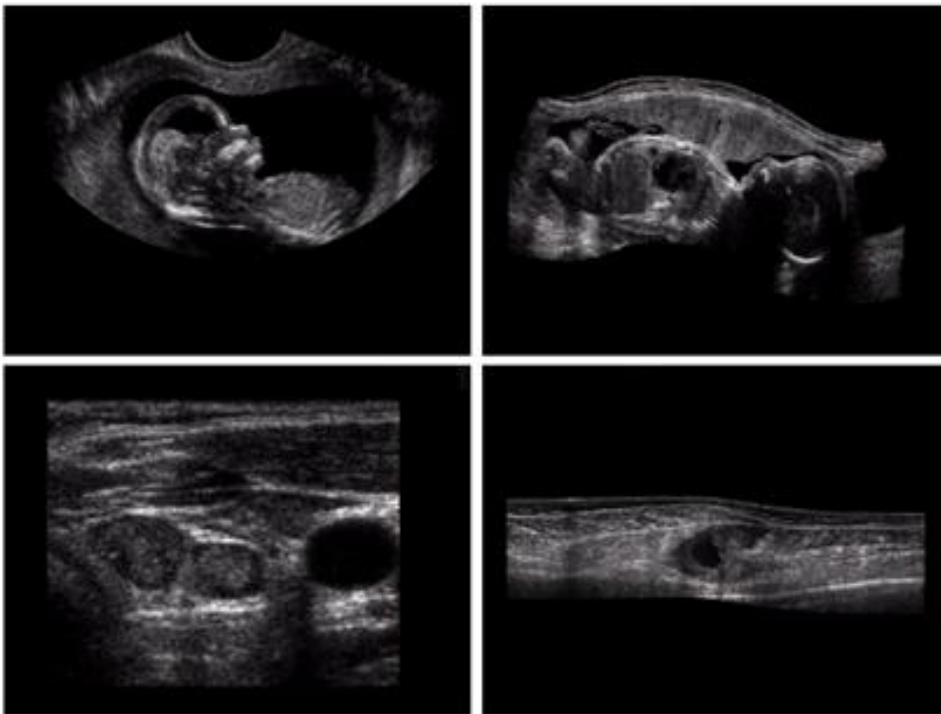
Cross section of 3D seismic model: arrow indicates bright spots for probable oil or gas trap.



Applications

High frequency (1-5 MHz, ultrasound) sound wave Imaging

Ultrasound travels through tissues and scatterings provides 3D rendition



A B
C D

A. Baby

B. Baby

C. Thyroids

D. Muscle



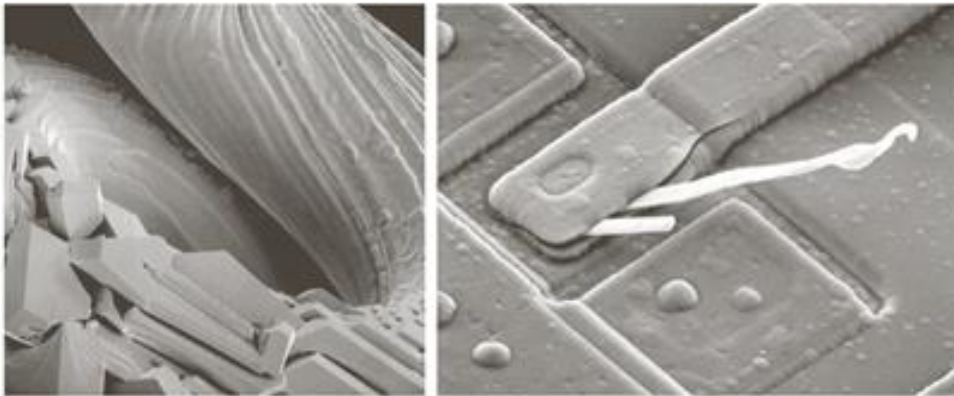
Applications

Electron Microscopy

Image formed by electron beams.

TEM: electrons transmitted through thin samples

SEM: bulky samples are scanned



A B A. SEM Tungsten

Filament

B. Damaged IC

Zoom of light microscopy - $1000\times$

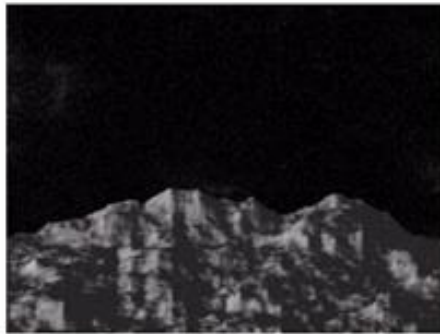
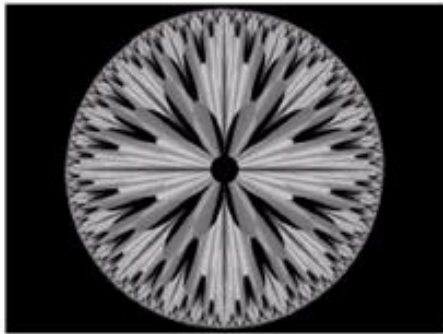
Zoom of electron microscopy – $10,000\times$



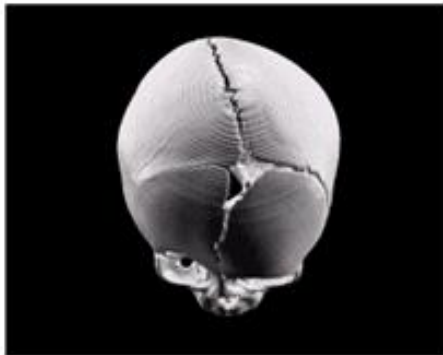
Applications

Synthetic Imaging

Used in fashion design, animation film industry



A B A. & B. Fractal Image

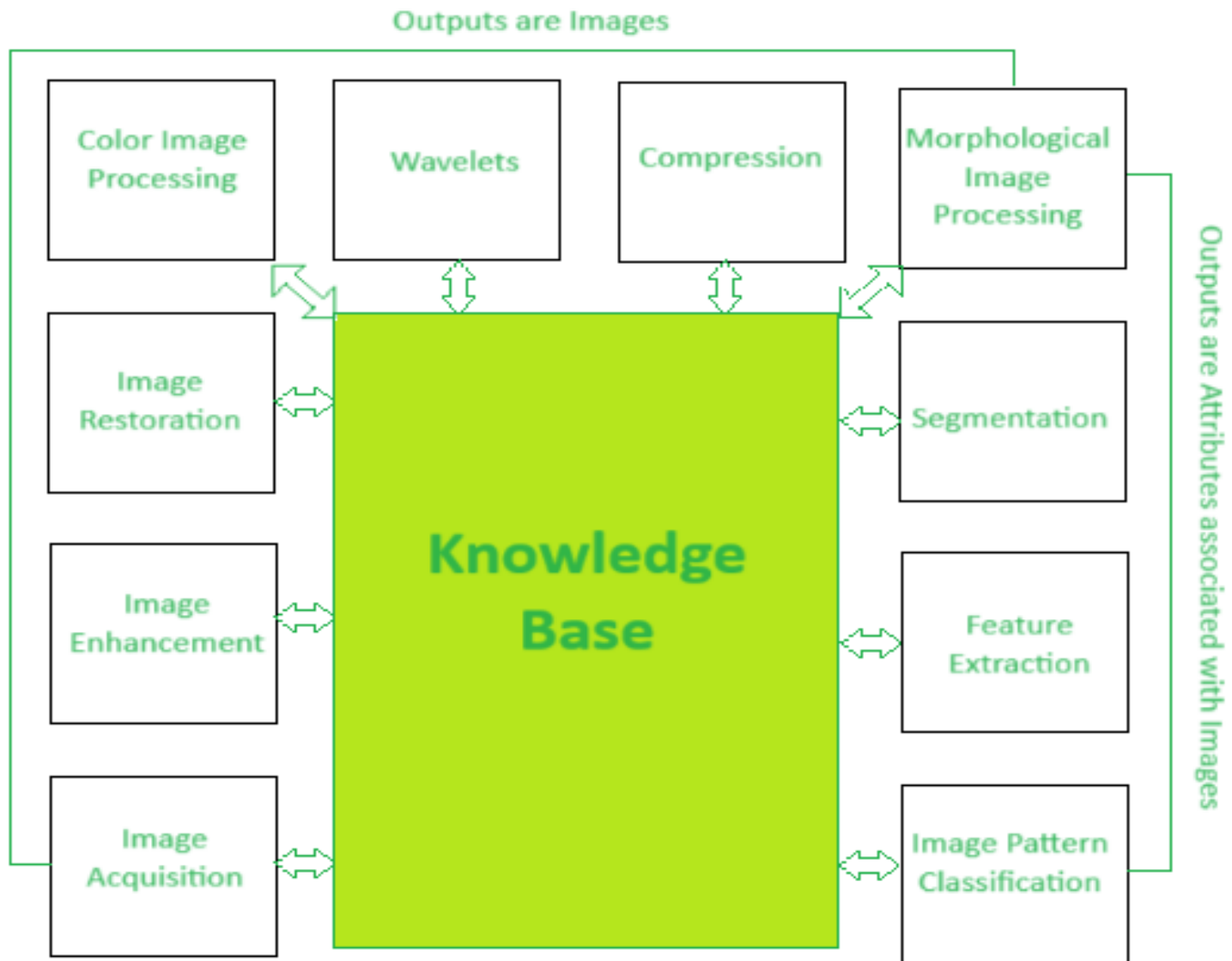


C D C. & D. 3D Objects

Fractal iteratively tiles: similar elements using a mathematical rule to design an object

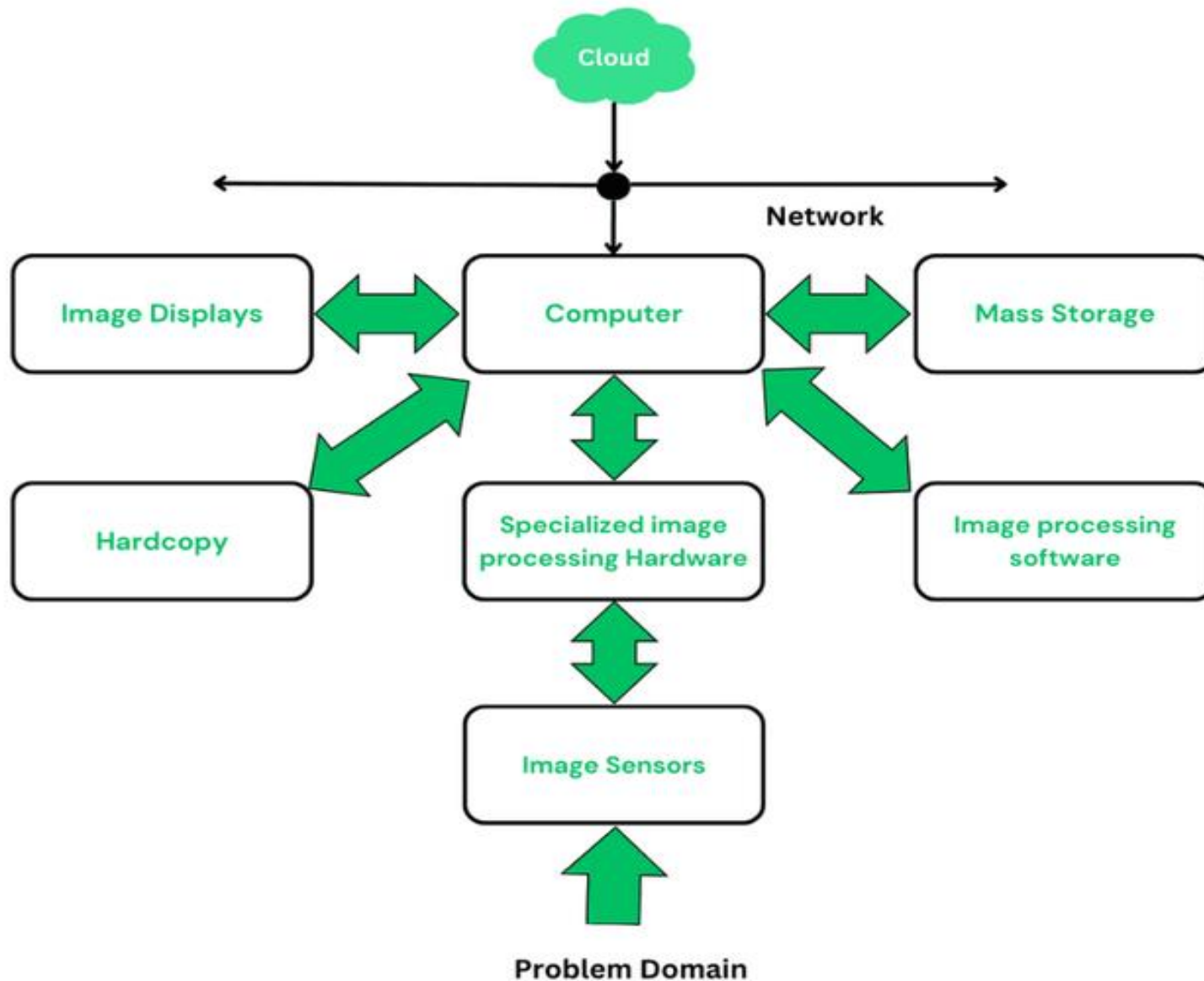


Approaches





Components





Components

Image Sensors: Image sensors sense the intensity, amplitude, co-ordinates and other features of the images and pass the result to the image processing hardware. It includes the problem domain.

Image Processing Hardware: Image processing hardware is the dedicated hardware that is used to process the instructions obtained from the image sensors. It passes the result to a general purpose computer.

Computer: Computer used in the image processing system is the general purpose computer that is used by us in our daily life.

Image Processing Software: Image processing software is the software that includes all the mechanisms and algorithms that are used in the image processing system.

Mass Storage: Mass storage stores the pixels of the images during the processing.

Hard Copy Device: Once the image is processed then it is stored in the hard copy device. It can be a pen drive or any external ROM device.

Image Display: It includes the monitor or display screen that displays the processed images.

Network: Network is the connection of all the above elements of the image processing system.



Questions???

Thank You!!!